

Early Warning Alarm System - final report

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List of Abbreviations

ADCS	Attitude Determination and Control System
CAN	Controller Area Network
CDH	Command and Data Handling (C&DH)
CDS	Cross-Domain Solution
CONOPS	Concept of Operations
COTS	Commercial Off-The-Shelf
EPS	Electrical Power System
EU	European Union
EWAS	Early Warning Alarm System
FMEA	Failure Modes and Effects Analysis
GEO	Geostationary Earth Orbit
GMAT	General Mission Analysis Tool
HIL	Hardware-in-the-Loop
IR	Infrared
ISL	Inter-Satellite Link
LEO	Low Earth Orbit
MEO	Medium Earth Orbit
MLI	Multi-Layer Insulation
MWIR	Medium-Wave Infrared
NATO	North Atlantic Treaty Organization
NFOV	Narrow Field of View
OBDH	On-Board Data Handling
RBAC	Role-Based Access Control
RF	Radio Frequency
SO	Strengths–Opportunities strategy
SWIR	Short-Wave Infrared
SWOT	Strengths, Weaknesses, Opportunities, and Threats
TCS	Thermal Control System
TTC	Telemetry, Tracking, and Command
UI	User Interface
VTS	Visualization Tool for Space Data
WBS	Work Breakdown Structure
WFOV	Wide Field of View
WT	Weaknesses–Threats strategy

1 Analysis

1.1 Mission Statement

There are increasing security concerns in the Baltic region, primarily associated with ongoing conflicts. Ballistic missile launches pose a significant risk to civilian populations due to insufficient warning times. Therefore, national security authorities require a wide-coverage, high-reliability detection system capable of persistent monitoring and timely alerting.

We propose a reliable space-based satellite system to detect missile launches at the earliest possible stage and to track them sufficiently enough to support rapid, preliminary impact-area estimation. The system will distribute warning information to national authorities responsible for national crisis management and emergency response. Additionally, the system should include the capability to store historical data of previous events to support threat analysis, enhance national resilience planning, and facilitate information sharing with partner governments and international crisis-response mechanisms.

1.2 Mission Definition

1.2.1 Mission Need and Purpose

The Early Warning Alarm System (EWAS) is defined as a space-based civil protection mission intended to provide national authorities responsible for national crisis management and emergency response with an early warning capability against ballistic missile launch threats. The mission is motivated by the limited warning time available during missile attacks and by the need to support rapid governmental and emergency-response decisions before a potential impact occurs.

The purpose of EWAS is to detect missile launch events at the earliest feasible stage, support preliminary assessment of the threat, estimate the potential impact area with an appropriate level of uncertainty, and deliver warning information to aforementioned authorities

EWAS is not intended to function as an offensive military system or as a weapon-control architecture. Its purpose is to provide timely warning and decision-support information for civil protection, national resilience, and emergency preparedness.

1.2.2 Mission Objectives

The EWAS mission objectives are to provide a reliable space-based capability that enables the detection, assessment and continuous improvement of missile early warning information.

The mission is intended to:

1. Detect ballistic missile launch events affecting the Baltic region as early as possible.
2. Track and process detected events sufficiently to support rapid preliminary estimation of the potential impact area.
3. Generate clear warning information for authorized national crisis-management and emergency-response authorities.
4. Maintain selected historical event data to improve future warning quality, support post-event analysis, refine classification methods, reduce false alarms, and calibrate preliminary impact-area estimation models.
5. Support national resilience planning and controlled information sharing with partner governments or international crisis-response mechanisms.

1.2.3 Mission Scope and Boundaries

The scope of the EWAS mission includes the space-based detection of missile launch events, preliminary event processing, initial threat assessment, warning generation, dissemination of warning products to authorized users, and storage of selected event data for technical and operational improvement.

Within this scope, EWAS should support:

- Launch detection and initial event confirmation;
- Preliminary classification of the detected event;
- Preliminary missile characterization based on observable event features and available aviation data;
- Preliminary trajectory and impact-area estimation;
- Secure delivery of warning information to national authorities responsible for national crisis management and emergency response;

- Post-event analysis and improvement of detection, classification, and estimation algorithms.

The mission boundaries are defined to preserve the civil protection character of the system. EWAS does not include missile interception, offensive targeting, weapon guidance, direct command of military response assets, final attribution of launch responsibility, or complete damage assessment. It also does not replace the national public alerting infrastructure; instead, it provides warning information to the authorities responsible for activating such systems.

1.2.4 Definition of Mission Success

Mission success for EWAS is defined by the system's ability to provide timely, reliable, and operationally useful missile warning information to national authorities responsible for civil protection and emergency response.

The mission may be considered successful if it can:

1. Detect relevant missile launch events within the defined surveillance region.
2. Generate a preliminary assessment of the event, including confidence level and estimated threat area.
3. Maintain sufficient reliability and availability to support operation during crisis conditions.
4. Provide warning products that are understandable and actionable for national emergency-management users without requiring them to operate or interpret the space system directly. Warning and data should be delivered within a time frame useful for crisis-management and civil-protection decisions.
5. Store data securely and use selected historical event data to support post-event review, missile characterization, reduction of false alarms, and refinement of detection, classification, and impact-area estimation methods.

2 Use Scenario

The primary use scenario of EWAS is missile threat detection and alert generation and dissemination. The flow of events in this scenario is generally linear and begins with the detection of a thermal change, which might indicate a missile launch, by autonomous satellite constellation. Such event serves as an entry point to a processing pipeline ending with alert dissemination as well as storing the event data for future uses.

Primary actors: Satellite operators, national authorities

Event flow:

1. Wide-field infrared sensor located aboard a GEO satellite detects a thermal spike and flags the event.
2. Onboard processor filters out noise and determines the source is a missile.
3. The two LEO satellites closest to the detected location orient their narrow-field sensors toward the moving target, perform stereoscopic tracking, and calculate a preliminary trajectory estimate.
4. The two LEO satellites downlink their encrypted telemetry and preliminary trajectory estimate to the ground segment, which verifies the data, calculates the definitive trajectory, and estimates the missile impact zone.
5. The system operator's console displays a warning and detailed information about the incoming missile and waits for confirmation.
6. Information about the missile event is written to a database after the operator's confirmation.
7. The system generates and disseminates adequate warnings to the authorities of endangered areas.

The flow of events in the use scenario is presented as the following use case diagram:

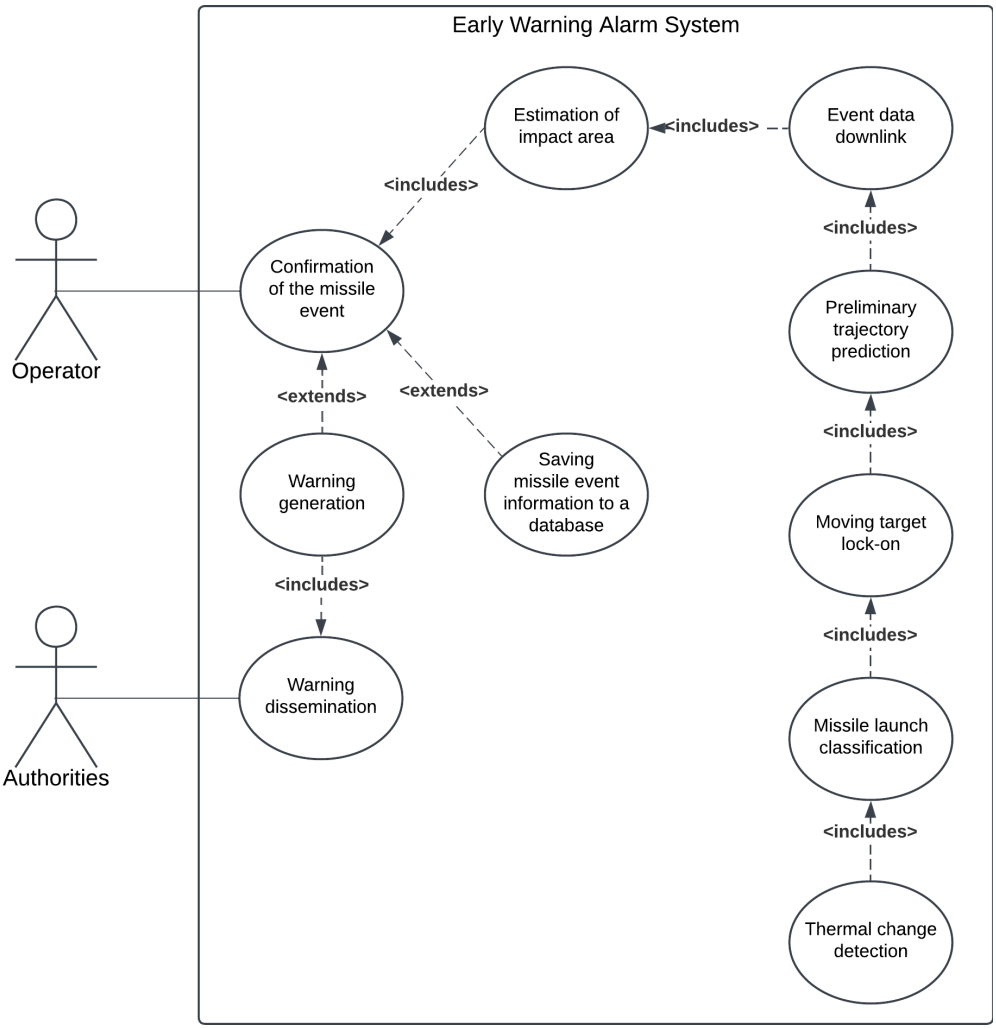


Figure 2.1: Use case diagram

3 Concept of Operations

The goal of this Concept of Operations is to provide a high-level description about what EWAS should achieve, to what extent, how it would be operated and what phases does the mission's life cycle consist of.

3.1 Mission objectives and requirements

The primary purpose of Early Warning Alarm System is to provide continuous surveillance to the Baltic region, detect ballistic missile threats through spikes in thermal signatures, estimate the potential impact zones and generate and broadcast alerts about the events.

Main objectives

- Detecting ballistic missile events threatening the Baltic region at the earliest possible stage of flight using infrared signatures.
- Tracking and prediction of detected missile's trajectory to efficiently estimate the possible impact area.
- Generating and dissemination clear warning information to authorities of potentially endangered areas enabling alerting civilians ahead of the impact.
- Storing historical event data to support analysis and improve the system's prediction capability.

Main requirements

- High global coverage
- Low end-to-end latency between the space and ground segments ensuring sufficient time for users to react to the incoming missile threat before impact.
- Accurate estimation of the preliminary impact area of the tracked missile.
- High classification accuracy resulting in maximizing the missile detection probability and minimizing the false alarm rate.
- High availability, preventing time windows in which the system is "blind" and ensuring proper alert dissemination.

3.2 System characteristics

From the perspective of the strategic operator, the EWAS acts as a seamless, automated "eye in the sky." The system abstracts complex orbital mechanics and raw sensor data into an intuitive, tactical display. Operators interact with a two-dimensional map on which the system automatically highlights the launch site, current position of the missile as well as estimated area of impact. Furthermore, the system displays color-coded alerts and features audio cues to eliminate human latency during a crisis. The system asks for cross-verification of such an event by the operator and if confirmed it generates and disseminates warnings about the incoming missile to adequate national authorities of endangered areas.

3.3 Mission operation

The Emergency Warning Alarm System operates in a continuous, synchronized loop consisting of surveillance, detection and tracking of detected missile threats.

Space segment operation

The space segment operates autonomously as a combination of GEO and LEO satellites to ensure high global coverage and high-resolution tracking. GEO satellites equipped with infrared sensors continuously scan the designated area to detect sudden thermal changes. When a potential missile threat is identified, the GEO segment cues the two LEO satellites closest to the detected location. The two LEO satellites track the missile from different observation angles, calculate a preliminary stereoscopic trajectory, and downlink the encrypted telemetry and preliminary trajectory estimate to the ground segment.

Ground segment operation

The ground segment receives the encrypted telemetry and preliminary trajectory estimate from the two LEO satellites, decrypts the data, and verifies its integrity. It then performs data fusion to calculate the definitive missile trajectory and estimate the potential impact area. Upon completion of the processing, it displays a warning to the operator, who verifies and confirms that a missile event took place. After confirmation, the system generates and disseminates clear warning information to the appropriate authorities in potentially endangered areas. Moreover, the ground segment stores selected historical event data to improve prediction capability and support post-event analysis.

3.4 Life cycle concept

1. **Concept definition**: The initial phase of the mission's life cycle consists of defining sensor parameters, designing the orbital constellation as well as preliminary testing

and software validation.

2. **Production and deployment**: Satellite components are built in cleanroom facilities, subjected to rigorous testing and integrated together. Afterwards, the satellites are launched into their designated orbits and the ground segment is established.

3. **Operation and maintenance**: Following the checkout and calibration the satellites enter their active service life during which trained operators monitor the state of the system and react accordingly to any detected missile threat.

   During this phase software upgrades are delivered to both the space and ground segments, while hardware maintenance is limited to the ground segment only.

4. **End-of-life and disposal**: At the end of operational life, to prevent space debris accumulation, LEO satellites are de-orbited and are left to burn-up safely in the Earth's atmosphere while GEO satellites are moved into graveyard orbit.

4 Proposed solution

4.1 Strategic Approach

The proposed solution addresses the critical security vulnerabilities of the Baltic region by fundamentally shifting the paradigm of early warning capabilities from a reactive, ground-based radar posture to a proactive, space-based observation framework. Ground-based systems are inherently limited by the curvature of the Earth, terrain masking, and fixed geographical positioning, which severely restricts their line-of-sight against low-altitude or highly compressed ballistic trajectories. This limitation manifests as the "insufficient warning times" currently challenging Baltic national security authorities.

To overcome these limitations, the concept relies on a hybrid satellite constellation composed of 6 GEO satellites and 72 LEO satellites deployed at an altitude of approximately 1,200 km and evenly distributed across 8 orbital planes with 9 satellites per plane to ensure continuous global coverage.[1] This specific architecture was selected over an alternative MEO/LEO framework based on a rigorous trade study. While MEO layers offer wider coverage outside the equatorial region, operating within the inner Van Allen radiation belt severely compromises sensor lifespans, drives up lifecycle costs and introduces critical detection gaps during initial launch sequences. The chosen GEO/LEO hybrid mitigates these risks. The GEO segment is primarily dedicated to detecting the initial booster phase of a missile launch, when the thermal signature is strongest and earliest detection is possible. The LEO segment then resolves polar coverage limitations and provides follow-on observation, enabling continuous tracking of the missile trajectory and supporting a more accurate preliminary impact-area estimation.

Rather than waiting for a ballistic missile to clear international borders or rise to a high altitude to be detected by ground radars, this concept focuses entirely on capturing the massive electromagnetic and thermal energy emitted during the initial ignition sequence (the boost phase). By detecting launch events as early as possible and maintaining uninterrupted tracking thereafter, the system is designed to provide national authorities with timely warning information for crisis management, civil protection, and emergency response. In addition, the system includes secure storage of historical event data to support post-event analysis, threat assessment, resilience planning, and information sharing with allied governments and international crisis-response mechanisms.

[1] The constellation sizing is justified by orbital mechanics simulations (GMAT) and coverage analysis (VTS Timeloop). The 72 LEO satellites at approximately 1,200 km altitude are distributed across 8 orbital planes (9 satellites each) so that at any given moment the constellation maintains uninterrupted visual coverage of the entire globe and no region falls outside sensor range between successive passes. The 6 GEO satellites are positioned to provide at least two independent geostationary vantage points over the primary area of interest, ensuring boost-phase detection and redundancy against single-satellite outages.

4.2 System Architecture

To realize this capability, the system is conceptually divided into three highly integrated, mutually supportive segments: the Space Segment, the Ground Segment, the User and Data Segment. Together, these segments form a fully automated pipeline that converts satellite observations into actionable warning products for national decision-makers.

4.3 The Space Segment

The Space Segment forms the core sensing layer of the system and consists of the proposed 6 GEO satellites and 72 LEO satellites. The GEO satellites provide persistent wide-area observation and are optimized for the earliest detection of missile launches during the boost phase, when the thermal plume is most intense and the launch location can be identified rapidly. As a GEO sensor detects an initial thermal burst, it performs preliminary onboard denoising to filter out solar bursts and terrestrial noise. If classified as a valid missile threat, the GEO satellite automatically transmits the target's precise coordinates to the two closest LEO satellites. Upon receiving these cross-cueing coordinates, these LEO satellites rapidly orient themselves toward the designated location to track the rising missile. By observing the target simultaneously, these two tracking satellites calculate a preliminary trajectory and transmit the encrypted telemetry data down to the Ground Segment.

This combined constellation ensures continuous surveillance, high temporal coverage, and resilient detection performance across the Baltic region and beyond. The sensors aboard the satellites are designed to detect transient infrared signatures associated with missile launches and to support automated cueing of follow-on observations.

4.4 The Ground Segment

The Ground Segment provides telemetry reception, mission control, and data processing. It is designed as a resilient and geographically distributed network of secure ground stations and processing facilities. Its primary function is to receive downlinked data from the Space Segment, process it with low latency, and transform raw observations into validated warning outputs.

Driven by the need for rapid and reliable performance, the ground infrastructure relies on automated processing pipelines and secure communications links. Upon receiving the encrypted telemetry data from the LEO layer, the Ground Station decrypts it and verifies data integrity. Thereafter, the ground architecture utilizes high-speed edge computing to calculate the definitive missile trajectory and projected impact zones while stripping away environmental background noise such as solar glint off the Baltic Sea, cloud reflections, or

industrial thermal outputs, leaving a clean, high-fidelity tracking signal. These systems are intended to ensure robust operation under contested or degraded conditions and to minimize the time between detection and dissemination of warning information.

4.5 The User and Data Segment

The final segment is the functional bridge to civilian and military decision-makers. It operates on a dual-pipeline concept:

The Immediate Alerting Pipeline: An automated, deterministic routing system that converts raw positional data into standardized air-defense and civil-protection formats, pushing immediate warning telemetry to national command centers.

The Telemetry Analytics Pipeline: A sovereign, highly encrypted data repository that securely stores every raw and processed dataset generated by the system for post-event analysis and strategic long-term planning.

4.6 Core Functional Pillars

The mission requirements are met through three distinct, conceptually separated functional operations within the system architecture.

Pillar I: Boost-Phase Infrared Detection The system's earliest warning capability is based on infrared detection of the missile's boost phase. The primary technical mechanism for achieving the "earliest possible" detection relies on IR sensing. During launch, ballistic missiles generate a strong thermal plume from burning propellants, producing a distinct infrared signature that can be detected from space. The GEO satellites are primarily responsible for this initial detection function, providing persistent observation while executing preliminary denoising and initiating rapid automated cueing of the tracking layer upon identifying a valid launch event.

Pillar II: Trajectory Tracking and Impact Estimation Once a launch is detected, the LEO constellation takes over the tracking function. The concept utilizes multi-satellite observation. When a launch event is triggered, adjacent satellites in the constellation cross-cue their sensors to observe the missile plume from multiple angles simultaneously. Specifically, the two closest LEO satellites orient toward the cross-cued coordinates, track the object to calculate a preliminary trajectory, and relay encrypted telemetry to the ground. Using multiple observation angles the satellites provide the geometric data needed to estimate the missile's trajectory with increasing confidence.

Pillar III: Low-Latency Dissemination and Sovereign Archiving To maximize warning value, the system is designed for rapid automated dissemination of validated information to national authorities. Warning messages are formatted at the ground station into lightweight, automated protocols that directly feed into national civil defense interfaces. Simultaneously, to address the mission's mandates for resilience planning and information sharing, all relevant data is securely archived to create a historical record of missile activity. This repository acts as a digital black box of regional airspace. Post-event, this data can be replayed to analyze adversary capabilities, refine air defense configurations, and improve national resilience structures.

5 Overall system requirements

This section defines the principal system-level requirements for the EWAS. The requirements are grouped by functional area and consolidated in Table 5.1 to support review, traceability, and verification planning.

Table 5.1: EWAS system requirements

ID	Requirement	Statement
Top-Level System Requirements		
R-001	Non-Offensive Operation	The system shall not provide targeting information for weapons systems, weapon guidance or fire-control data, interceptor command-and-control functions, or direct control of military response assets.
R-002	Scope Limitation	The system shall not be required to provide definitive missile-type identification, final warhead characterization, exact explosive-yield estimation, or comprehensive post-impact damage assessment.
R-003	Authorized User Access	System outputs shall be accessible only to authorized governmental and institutional users, including crisis-management and civil-protection authorities.
R-004	Data Usage	Historical mission data shall be used solely for algorithm improvement, system calibration, performance verification, operator training, and post-event technical analysis.
Constellation and Coverage Requirements		
R-005	Space Segment Architecture	The EWAS space segment shall consist of six GEO satellites and 72 LEO satellites operating at an altitude of 1,200 km.
R-006	Continuous Coverage	The constellation shall provide continuous surveillance of the designated monitoring region on a 24-hour basis.
R-007	Boost-Phase Detection Coverage	The orbital architecture shall provide observation capability during the missile boost phase to enable the earliest feasible launch detection.
R-008	Coverage Availability	The constellation shall maintain continuous temporal coverage of the designated surveillance region without operational gaps that would prevent launch detection.
Performance Requirements		
R-009	Launch Detection	The system shall detect ballistic missile launch events by observing characteristic infrared signatures generated by missile propulsion systems.
R-010	Probability of Detection	The probability of detection shall exceed 95% for missile threats defined within the mission scope.
R-011	End-to-End Latency	The total elapsed time from initial detection to delivery of a warning product to authorized users shall not exceed 60 seconds.
R-012	Preliminary Impact Estimation	Following launch detection, the ground segment shall generate an initial impact-area estimate using the preliminary trajectory and telemetry downlinked by the two tracking LEO satellites.

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ID	Requirement	Statement
R-013	Uncertainty Quantification	Impact-area estimates shall be presented as probabilistic confidence regions with clearly defined confidence levels and uncertainty boundaries.
R-014	Dynamic Update Capability	The ground segment shall continuously refine the definitive trajectory and impact-area estimates as additional tracking data become available.
R-015	Operational Availability	The overall system availability shall exceed 99.9%.
R-016	Fault Tolerance	The loss of any single satellite, ground station, or communication node shall not result in loss of mission capability.
R-017	System Redundancy	The system architecture shall incorporate sufficient redundancy to maintain operational performance during component failures.
R-018	False Alarm Rate	The false alarm rate shall not exceed one false warning event per month per monitored sector.
R-019	Event Classification	The system shall automatically distinguish missile launch events from non-threat infrared sources through onboard and ground-based processing.
Space Segment Requirements		
R-020	Infrared Sensor Payload	Each satellite shall carry infrared sensing equipment capable of detecting missile-plume signatures against background environmental conditions.
R-021	Sensor Coverage	The payload shall support wide-area surveillance while maintaining sufficient sensitivity to satisfy launch-detection requirements.
R-022	Onboard Event Processing	Satellites shall perform onboard event filtering and preliminary event characterization to reduce communication latency and bandwidth requirements.
R-023	Preliminary Trajectory Estimation	The onboard processing subsystems of the two cued LEO satellites shall use stereoscopic observations to calculate a preliminary trajectory estimate before transmitting it to the ground segment.
R-024	Pointing Stability	The spacecraft shall maintain pointing accuracy and stability sufficient to achieve mission detection-performance requirements.
R-025	Stray-Light Mitigation	The spacecraft design shall minimize stray-light contamination that could degrade sensor performance.
Ground Segment Requirements		
R-026	Data Fusion	The ground segment shall support real-time fusion of observations from all GEO and LEO satellites.
R-027	Threat Assessment Processing	The ground segment shall fuse the LEO telemetry and preliminary trajectory estimate to calculate the definitive missile trajectory and impact-area prediction in real time.
R-028	Decision Support Interface	The user interface shall visualize the launch location, estimated trajectory, predicted impact area, confidence level, and uncertainty region.

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Table 5.1 continued from the previous page

ID	Requirement	Statement
R-029	Human Factors Design	The interface shall be designed to minimize operator misinterpretation during high-stress situations.
R-030	Command and Control	The ground segment shall provide redundant command-and-control capabilities for constellation management.
Communications Requirements		
R-031	Secure Communications	All communications shall employ encryption mechanisms compliant with applicable national security requirements.
R-032	Anti-Jamming Protection	Communication links shall incorporate protection against jamming, spoofing, and unauthorized access.
R-033	Near-Real-Time Data Transmission	The communication subsystem shall support near-real-time transmission of sensor and telemetry data between the space and ground segments.
Data Management and Security Requirements		
R-034	Role-Based Access Control	The system shall implement Role-Based Access Control (RBAC) to restrict access to authorized users.
R-035	Secure Data Storage	All archived mission data shall be stored in encrypted repositories.
R-036	Data Retention	Historical mission data shall be retained for a minimum period of five years.
R-037	Audit Logging	The system shall maintain audit records of user actions, system events, and warning-dissemination activities.
Verification Requirements		
R-038	Simulation-Based Verification	System performance shall be verified through representative simulation campaigns covering a range of missile-launch scenarios and environmental conditions.
R-039	Hardware-in-the-Loop Testing	Critical sensing and processing functions shall undergo Hardware-in-the-Loop (HIL) testing prior to operational deployment.
R-040	Operational Validation	End-to-end operational exercises shall demonstrate compliance with detection, latency, availability, and warning-dissemination requirements.
R-041	Performance Monitoring	Archived operational data shall be periodically analyzed to support algorithm refinement and continuous system improvement.

6 Functional Analysis

Functional analysis defines how EWAS performs its mission from initial observation to warning delivery and long-term system improvement. It describes the main operational flows between the GEO and LEO satellite layers, the ground segment, and the national authority users. The purpose of this section is to clarify the sequence of functions required to detect a potential missile launch, process and validate the event, estimate the threat trajectory and impact area, disseminate an alert, and maintain the system during routine and crisis operations.

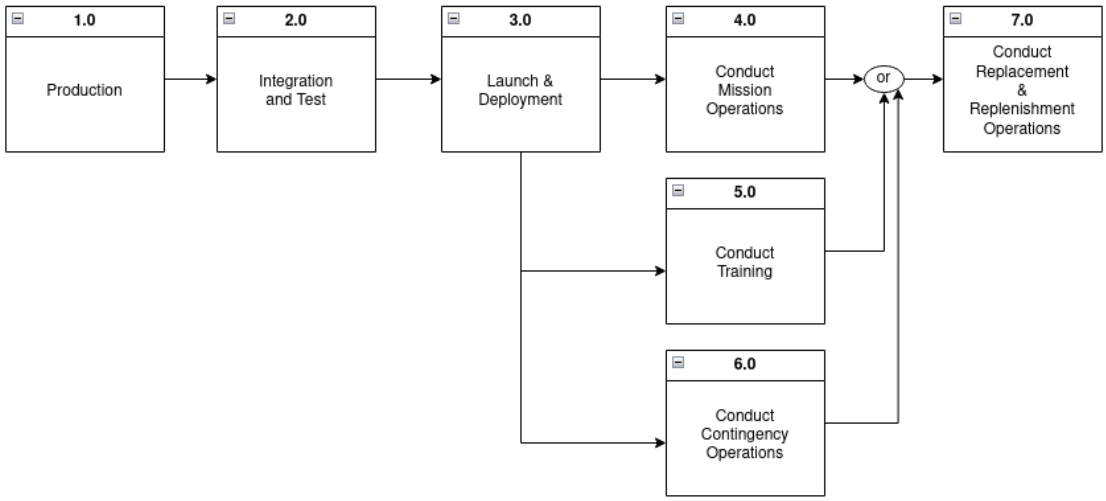


Figure 6.1: Top Level Flow

6.1 Primary Threat Detection and Warning Flow

The primary responsibility of the EWAS mission is to detect missile threats worldwide, determine whether they could affect the Baltic region, and generate dedicated alerts for emergency authorities. This mission is conducted in accordance with the functional flow shown in Figure 6.2.

The combined constellation of six GEO satellites and 72 high-inclination LEO satellites is responsible for detecting unusual thermal bursts. GEO satellites provide persistent wide-area observation where viewing geometry is favorable, while LEO satellites provide high-elevation coverage at high latitudes. The detecting satellite conducts denoising to rule out solar bursts, Earth background noise, and other non-threat thermal signatures.

In case of an unusual burst detection, the burst coordinates are sent to the two LEO satellites closest to the detected location. The two LEO satellites track the potential missile

from different observation angles and calculate a preliminary stereoscopic trajectory. They then transmit the encrypted telemetry and preliminary trajectory estimate to the Ground Segment.

Finally, the Ground Segment decrypts the received telemetry data and confirms its integrity. It then fuses the observations to calculate the definitive trajectory and estimated impact zone. The system generates an alert, which is then sent to the appropriate national authority body.

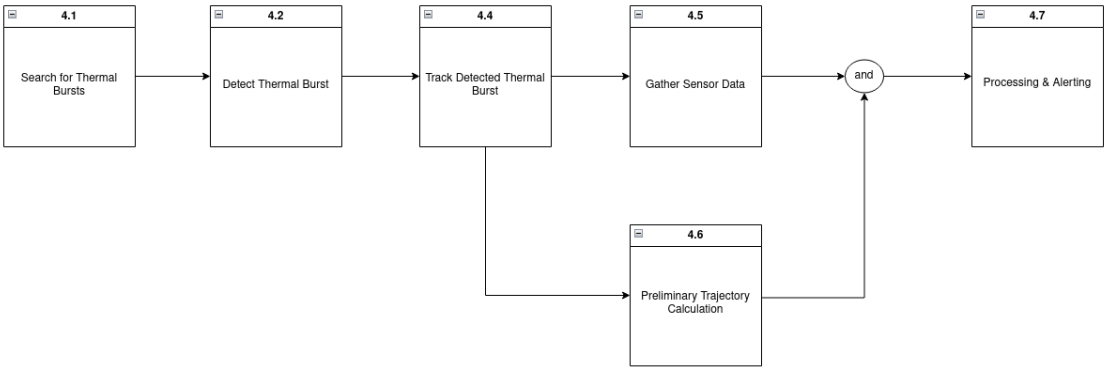


Figure 6.2: Second Level Flow

6.2 Data Management and Continuous Improvement Flow

This flow operates asynchronously to the primary warning flow, fulfilling the mission's objective to use historical data for system refinement and national resilience.

- Archive all raw sensor readings, onboard processing logs, telemetry records, and final generated warning outputs in encrypted system storage.
- Retrieve historical event data for authorized post-event technical analysis, lessons-learned review, and missile characterization.
- Use archived data to refine detection, classification, denoising, tracking, and impact-area estimation algorithms.
- Reduce false alarm rates by comparing confirmed missile events with non-threat thermal signatures such as solar bursts and Earth background noise.
- Prepare validated software and algorithm updates for the space segment and ground processing center.
- Securely uplink approved updates to satellites and deploy them to the ground segment during planned operation and maintenance windows.
- Share sanitized historical event data with partner governments and international crisis-response mechanisms when authorized.

6.3 System Maintenance and Support Flow

This background operational loop ensures that EWAS maintains high availability, reliable coverage, and stable performance during both routine operation and crisis conditions.

- Continuously monitor satellite health and status telemetry from the space segment.
- Verify communication links between GEO satellites, LEO satellites, ground stations, and national authority interfaces.
- Perform attitude and orbit control to maintain required satellite pointing, orbital positioning, and coverage of the monitored region.
- Regulate spacecraft power and thermal conditions using the Electrical Power System (EPS) and Thermal Control System (TCS).
- Detect subsystem anomalies and initiate fault-management procedures when degraded performance is observed.
- Maintain ground segment hardware, processing infrastructure, databases, and operator consoles.
- Apply tested software patches and configuration updates to preserve redundant and uninterrupted operations.

7 Subsystem Design

The purpose of this chapter is to define the architecture of the Early Warning Alarm System space segment. Following the top-level mission objectives and requirements, the spacecraft architecture is decomposed into specialized operational subsystems. To ensure continuous, 24/7 global temporal coverage with minimal blind spots and meet the strict latency requirements, the design focuses on a modular architecture. This approach couples a standardized satellite bus with highly specialized mission payloads optimized for multi-tier ballistic event tracking.

To fulfill the EWAS mandate of global, high-coverage civil protection, the system architecture utilizes a dual-tier space segment constellation:

- **High-Altitude Detection Layer (GEO):** Comprises 6 satellites in Geosynchronous Earth Orbit (35,786 km), spaced uniformly at 60 degree intervals along the equator to provide continuous staring detection across mid-and-low latitudes.
- **Low-Altitude Tracking Layer (LEO):** Comprises a Walker Delta constellation of 72 small satellites operating at an altitude of 1,200 km across 8 high-inclination orbital planes, with 9 satellites per plane. This configuration ensures complete high-latitude and polar coverage with dense multi-satellite stereoscopic overlap.

The subsystem design presented in this section prioritizes ultra-high-sensitivity detection, structural longevity, and rapid, secure data dissemination.

7.1 Standard Satellite Bus Architecture

The EWAS space segment is based on a modular satellite bus utilizing flight-proven Commercial Off-The-Shelf (COTS) components to minimize development risks, control baseline procurement costs, and ensure maximum reliability. This standardized platform provides the essential operational infrastructure via the following subsystems:

Structure: Modular aluminium frame designed for launch load endurance and internal component protection. It incorporates localized structural shielding to mitigate radiation degradation (radiation hardening) encountered during extended operations in GEO and persistent LEO orbital environments.

Electrical Power System: Deployable solar arrays and high-capacity Lithium-Ion (Li-Ion) battery storage to guarantee uninterrupted power distribution during orbital eclipse phases. The system is scaled to accommodate standard spacecraft operations alongside the transient peak-power demands of the primary computational assets during active threat tracking.

Thermal Control System: Passive thermal regulation utilizing Multi-Layer Insulation (MLI), optical surface reflectors, and dedicated heat-pipe radiators. Special architectural isolation is implemented to handle the cryogenic cooling requirements of the infrared sensors to maintain nominal detector sensitivity.

Propulsion System: A low-thrust monopropellant or electric propulsion module. Its primary functions are to perform autonomous station-keeping manoeuvres to maintain precise slot positioning within the GEO/LEO constellation grids, and execute orbital phase management.

Communications Subsystem: Incorporates high-gain, highly directional antennas coupled with a dedicated hardware cryptographic module to safeguard data integrity. Crucially, the system features Inter-Satellite Links (ISL) to route data across the orbital shells, allowing a satellite tracking an event over remote oceans to relay alert data to a node directly over a secure Baltic ground station, preserving the under-60-second end-to-end latency constraint.

7.1.1 GEO vs LEO Platform Differentiation

While both platforms share COTS-derived design patterns, their drastically different orbital environments require separate engineering margins. Table 7.1 outlines the specific requirements allocated to the GEO and LEO satellite buses.

Table 7.1: Engineering Requirements for GEO and LEO Satellite Bus Configurations

Subsystem	GEO Bus Specifications	LEO Bus Specifications
Structure	Heavy radiation shielding for Geostationary environment.	Lightweight composite/aluminium structure optimized for batch launch.
EPS	Large solar arrays, continuous exposure with rare equinox eclipses.	High-cycle Li-Ion batteries to handle 15 orbital eclipses per day.
ADCS	Earth-pointing nadir stabilization with low angular velocity.	High-agility reaction wheels and magnetorquers for fast target cross-cueing.
TTC / ISL	Persistent high-power ground links and multi-channel ISL routing nodes.	Dynamic, steerable ISL antennas to lock onto moving adjacent LEO nodes.

7.2 Mission-Specific Payloads

Global Multi-Tier Infrared (IR) Sensor Suite: The primary mission payload is an advanced infrared optical assembly tailored to capture the massive thermal energy of ballistic missile plumes during the boost phase against the cold background of space. The

sensors operate in complementary Short-Wave (SWIR) and Medium-Wave (MWIR) bands to minimize false alarms and maximize detection probability.

GEO Observation Layer: Focuses on global detection using Wide Field of View (WFOV) staring optics. While spatial resolution is low (approx. 1–2 km per pixel), the extreme radiometric contrast of a rocket motor firing acts as an immediate point-source digital trigger.

LEO Tracking Layer: Features Narrow Field of View (NFOV) tracking sensors integrated onto dynamic, dual-axis gimbals. Cued by initial high-altitude alerts, these optics mechanically steer their line of sight to lock onto the target footprint, crossing paths with adjacent LEO nodes to establish multi-angle stereoscopic tracking of the missile post-burnout.

Key Design Drivers: High imagery sampling rates for rapid trajectory tracking and precise signal discrimination to isolate missile plumes from natural solar glints or heavy background atmospheric clutter.

7.3 On-Board Data Handling and Advanced Processing

The On-Board Data Handling (OBDH) system manages all data flows and executes mission-critical software. EWAS utilizes a high-performance computational architecture to process information in real-time while on orbit. To handle data transmission reliably under extreme environmental conditions, the internal data architecture implements the **SpaceWire** protocol for high-rate payload data and a redundant **CAN bus** for critical platform telemetry.

Key design drivers include:

- **On-board Event Filtering:** Capability to distinguish between missile plumes and background noise using advanced algorithms to reduce false alarm rates.
- **Latency Optimization:** Significant reduction in warning time by generating detection alerts locally, eliminating the need to downlink massive volumes of raw infrared imagery.
- **Secure Data Management:** All processed data and event logs are encrypted and archived in a secure on-board memory unit for later technical review and algorithm refinement.

7.4 Attitude Determination and Control System

Given the high velocity of LEO nodes and the dynamic motion of GEO platforms relative to sudden threats, the Attitude Determination and Control System (ADCS) must continuously manipulate the spacecraft's spatial orientation to maintain stable ground tracking.

Key design drivers include:

- **Precision Pointing:** System-level capability to aim the IR optics at targeted coordinates without image blurring. The target pointing stability is maintained below 2 arcseconds per second to ensure clear pixel capture.
- **Stability Management:** Active elimination of high-frequency micro-vibrations and structural jitter that confuse small plume detection algorithms.
- **Autonomous Target Tracking:** Dynamic adjustments to the spacecraft attitude in real-time to match cued geographic coordinates received from the detection layer.

7.5 Spacecraft Subsystem Interactivity and Command Data Flow

To meet the strict under-60-second end-to-end latency constraint, the command logic and cross-layer data exchange operate as a synchronized, autonomous loop across the space and ground segments. The operational data flow is executed as follows:

1. **GEO Constellation Detection:**
    - The GEO WFOV infrared payload detects a sudden thermal burst within the monitored region.
    - The onboard OBDH processing layer conducts real-time digital denoising to filter out solar bursts, solar glints, and Earth background noise.
    - The onboard classification algorithm validates the signature as a potential missile and immediately commands the TTC system to transmit the target's coordinates to the 2 closest LEO satellites via Inter-Satellite Links (ISL).
2. **LEO Constellation Tracking:**
    - The 2 cued LEO satellites receive the threat coordinates via their ISL receiver terminals.
    - The LEO ADCS autonomously re-orients the satellite bus, and the dual-axis gimbals mechanically lock the NFOV tracking sensors onto the moving missile plume.

- The cued LEO satellites track the target simultaneously from different observation angles, allowing the OBDH layer to calculate a preliminary stereoscopic trajectory.
- The LEO communications subsystem encrypts the telemetry and trajectory dataset via the hardware cryptographic module and downlinks it to the Ground Station Network.

3. **Ground Segment Processing:**

- The Ground Station receives the encrypted telemetry packet, decrypts it, and verifies its data integrity.
- High-speed data fusion pipelines calculate the definitive missile trajectory and predict the civil impact zones.
- The ground system automatically generates a live, standardized alert and routes it directly to the appropriate national authority body for civil protection.

8 Stakeholder Analysis

The primary objective of this stakeholder analysis is to identify all key actors, evaluate their power (influence) and interest in the project, and establish tailored management strategies to ensure seamless cooperation, secure data workflows, and mitigate political or operational resistance.

We have identified the stakeholders and divided them into four primary groups. Each group contains specific organizations, agencies, or team roles that directly interact with the system.

8.1 Institutional Users & Decision Makers

This group consists of high-level government entities and civil protection agencies that rely directly on EWAS outputs to protect human life and manage national crises. They do not operate the satellites, but they are the primary consumers of the warning products.

- **National Crisis Management & Emergency Response Authorities:** Specific regional government security centers and sovereign security authorities within the alliance's territory, alongside their respective Ministries of the Interior. These institutions require immediate, clear data to initiate national safety protocols.
- **Emergency Warning Centers:** Operational hubs (such as civil defense headquarters and regional emergency dispatch centers) that receive the 2D tactical maps and probabilistic impact ellipses to coordinate local actions.
- **Designated Government Decision-Makers:** Ministers of Defense, Ministers of Interior, and heads of state who need high-fidelity data to make rapid geopolitical and defensive decisions during a missile crisis.

8.2 Operational & Technical Team

This group includes the internal team roles responsible for the day-to-day survival, maintenance, and continuous optimization of both the Space and Ground segments. They ensure the system maintains its required 99.9% availability.

- **System Operators & Satellite Controllers:** Tactical personnel working in a secure ground station. All tracking data from 72 LEO satellites across 8 orbital planes is routed via 6 GEO relays using Inter-Satellite Links (ISL).

- **Subsystem & Payload Engineers:** Responsible for satellite bus configuration, multi-orbit mechanics, fleet synchronization (6 GEO + 72 LEO), cryogenic IR sensor systems, and ISL routing protocols.
- **Data Analysts & Software Developers:** Manage telemetry analytics pipeline, secure data warehouse, trajectory modeling, and multi-satellite data fusion algorithms.

8.3 Industrial Suppliers & Enablers

These actors provide infrastructure, hardware, and logistics for deployment.

- **Foreign Launch Service Providers:** Companies and regulators responsible for deploying LEO and GEO satellites into orbit.
- **Commercial Off-The-Shelf (COTS) Manufacturers:** Suppliers of standardized satellite components such as solar arrays, structures, and processors.
- **Specialized ISL & Optical Payload Contractors:** Provide SWIR/MWIR sensors and laser/RF inter-satellite communication terminals.

8.4 External Partners & Alliances

This group includes international stakeholders affected by the system.

- **Partner Governments:** Allied nations receiving sanitized threat data via Cross-Domain Solutions (CDS).
- **International Alliances & Funding Bodies:** Provide funding and require milestone tracking and stability outcomes.
- **Space Regulatory Agencies & Traffic Management:** Manage orbital slots, RF allocation, and debris mitigation (LEO + GEO end-of-life operations).
- **General Public / Civilians:** Indirect beneficiaries of early warning infrastructure.

8.5 Detailed Analysis

The following table presents a detailed stakeholder analysis, identifying their respective levels of power and interest, key expectations and requirements, and the proposed management strategy for each stakeholder group.

Table 8.1:
Stakeholder Analysis and Management Strategy

ID	Stakeholder	Power	Interest	Core Expectations & Requirements	Management Strategy
1	National Crisis Management Authorities	High	High	Require immediate, clear data to activate national safety protocols and coordinate civil protection across the Baltic states.	**Manage Closely:** Conduct regular high-level operational reviews and integrate their communication protocols early.
2	Emergency Warning Centers	High	High	Require end-to-end latency below 60 seconds and intuitive 2D tactical maps showing probabilistic impact ellipses.	**Manage Closely:** Involve them deeply in the User Interface (UI) design phase to minimize human response latency during crisis conditions.
3	Designated Government Decision-Makers	High	High	Expect highly reliable decision-support information and strict assurance that EWAS remains a non-offensive, defensive civil asset.	**Manage Closely:** Provide high-level technical updates emphasizing compliance with civil protection boundaries.
4	System Operators & Satellite Controllers	High	High	Require 99.9% system availability, high-fidelity visualization tools, and a streamlined single-station ground interface receiving a continuous data stream routed from the 72 LEO satellites via the 6 GEO relays.	**Manage Closely:** Include them in HIL simulations that test data handovers and routing between the shifting LEO and fixed GEO layers.
5	Subsystem & Payload Engineers	Low	High	Require stable requirements to manage multi-orbit fleet synchronization (6 GEO + 72 LEO across 8 orbital planes), complex tracking geometry from 1,200 km, and structural radiation shielding.	**Keep Informed:** Establish clear communication channels between technical layers to ensure seamless hardware and dual-orbit integration.

Continued on next page

Table 8.1 – continued from previous page

ID	Stakeholder	Power	Interest	Core Expectations & Requirements	Management Strategy
6	Data Analysts & Software Developers	Low	High	Require access to the secure data warehouse to refine algorithms capable of fusing overlapping trajectory data from multiple LEO satellites simultaneously.	**Keep Informed:** Provide structured access to post-event data logs to support continuous software refinement and false-alarm reduction.
7	Foreign Launch Service Providers	High	Low	Demand strict compliance with launch vehicle safety regulations, payload dimensions, and export control laws.	**Keep Satisfied:** Handle as a major supply chain risk. Secure complex, multi-launch contracts needed to deploy 72 LEO satellites across 8 orbital planes, likely through multiple rideshare launches, and 6 heavy GEO satellites.
8	COTS Manufacturers	Low	High	Require clear technical interface specifications and timely milestone payments for standardized, mass-produced components to support the 72 satellite LEO fleet.	**Keep Informed:** Maintain strict quality assurance monitoring during the high-volume component procurement and integration phases.
9	Specialized ISL & Optical Payload Contractors	Low	High	Require precise technical profiles to manufacture 72 sets of high-speed tracking telescopes (SWIR/MWIR) and reliable, dynamic laser/RF Inter-Satellite Link terminals.	**Keep Informed:** Maintain intense engineering coordination to ensure rapid data cross-linking between fast-moving LEOs and stationary GEO networks.
10	Baltic Partner Governments	Low	High	Expect controlled access to threat telemetry and launch archives to enhance their own national resilience and regional security planning.	**Keep Informed:** Utilize secure Cross-Domain Solutions (CDS) to safely share sanitized data without exposing core system vulnerabilities.

Continued on next page

Table 8.1 – continued from previous page

ID	Stakeholder	Power	Interest	Core Expectations & Requirements	Management Strategy
11	International Alliances & Funding Bodies (NATO / EU)	Low	High	Expect efficient use of dual-use resources, strict project milestone tracking, and clear alignment with regional civil protection frameworks.	**Keep Informed:** Provide periodic progress reports and financial audits to secure long-term institutional backing.
12	Space Regulatory Agencies & Traffic Management	High	Low	Demand compliance with radio frequency allocations for dense ISL networks, safe positioning within crowded GEO slots, and active space debris mitigation at 1200 km (LEO end-of-life disposal).	**Keep Satisfied:** Submit complex orbital slot applications and regulatory frequency coordination paperwork very early in the system design phase.
13	General Public / Civilians	Low	Low	Expect general national safety and infrastructure resilience without direct operational awareness of the satellite layers.	**Monitor:** Track broader public trust metrics in national defense warning infrastructures; no direct communication or marketing needed.

8.6 Stakeholder Matrix

Figure 8.1 illustrates the distribution of stakeholders across the power–interest matrix according to their respective levels of power and interest.

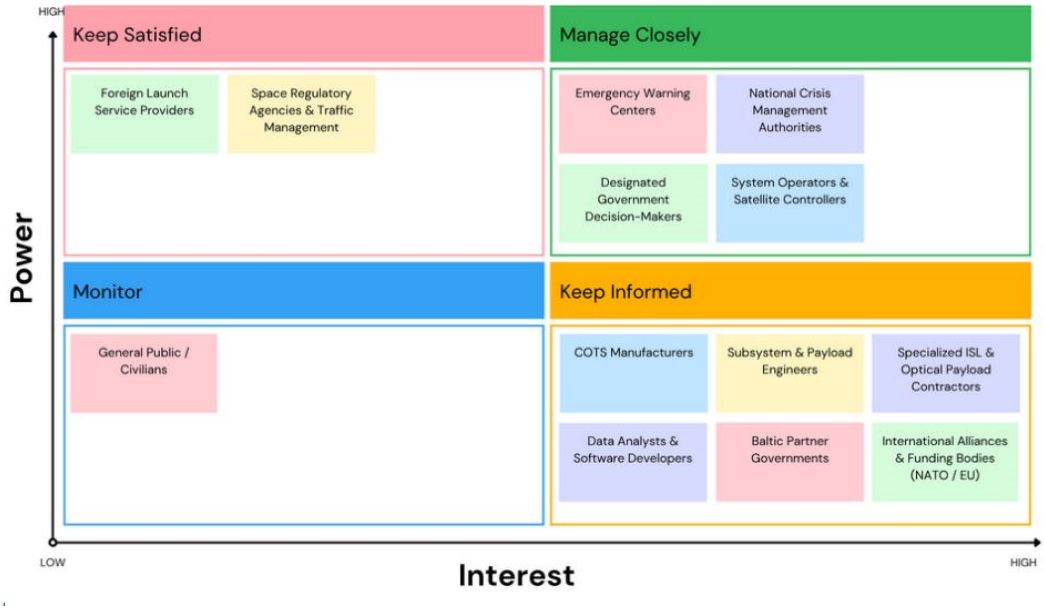


Figure 8.1: Stakeholder power–interest matrix.

8.7 Stakeholder Management Synthesis

The stakeholder analysis highlights four main operational focus areas:

- **Manage Closely (High Power, High Interest):** Institutional users and operators requiring low-latency decision support and intuitive interfaces during crisis conditions.
- **Keep Informed (Low Power, High Interest):** Engineers, developers, and suppliers ensuring continuous system development and integration across LEO and GEO layers.
- **Keep Satisfied (High Power, Low Interest):** Launch providers and regulatory agencies requiring compliance and early coordination.
- **Monitor (Low Power, Low Interest):** General public, with focus on long-term trust in national warning infrastructure.

9 Risk/SWOT Analysis

The conducted SWOT analysis indicates that the proposed satellite-based Early Warning Alarm System occupies a strong strategic position due to its unique operational capabilities and its alignment with the growing demand for civil protection and national resilience in the Baltic region.

9.1 Strengths

The most significant strengths identified are the high reliability of the satellite constellation and the absence of a dedicated regional early-warning capability offering comparable functionality. These factors achieved the highest weighted scores, demonstrating that the proposed system addresses a critical capability gap while providing a reliable and innovative solution for missile early warning. Early threat detection and high system availability further enhance the mission's operational value, while the capability to store historical event data supports post-event analysis, continuous improvement of detection algorithms, reduction of false alarms, and refinement of future warning products.

9.2 Opportunities

The identified opportunities strongly complement these internal strengths. The most important external factors include the increasing demand for effective civil protection and missile early-warning capabilities in the Baltic region, together with the availability of European Union and NATO programmes supporting resilience, crisis management, and dual-use technologies.

The interaction analysis demonstrates that the project's strengths effectively enable the exploitation of these opportunities. In particular, the combination of high system reliability, continuous monitoring capability, and early-warning functionality creates favourable conditions for institutional cooperation, long-term programme support, and integration into future regional civil protection and resilience frameworks. Furthermore, the potential expansion of the satellite constellation provides additional long-term development opportunities, although its implementation will depend on technical feasibility and the availability of financial resources.

9.3 Weaknesses

Despite these advantages, several internal weaknesses may reduce the project's ability to fully capitalize on emerging opportunities. The most significant weakness is the high cost of satellite deployment and long-term operations, which directly affects the feasibility of future constellation expansion and the development of the mission's ground segment.

Additional challenges include gradual satellite degradation throughout the mission lifetime, limited maintenance possibilities in orbit, and the complexity of integrating observations from multiple satellites into a coherent monitoring system. False-positive detections caused by atmospheric phenomena may also reduce operational efficiency if filtering and classification methods are not continuously improved. Overall, these weaknesses primarily affect the scalability and long-term sustainability of the system rather than its core operational objectives.

9.4 Threats

The analysis also identifies several external threats that could negatively influence the long-term success of the mission. The most significant risks are cyberattacks and anti-satellite activities, including jamming and spoofing, which received the highest weighted importance among the identified threats.

Since the mission relies on continuous satellite operation, secure communications, and reliable data transmission between the space and ground segments, maintaining robust cybersecurity and resilience against electronic interference is essential. An additional critical vulnerability is the dependence on Inter-Satellite Links connecting the LEO tracking satellites with the GEO communication and early-warning satellites. Disruption, jamming, or spoofing of these links could significantly degrade the system's ability to transmit detection data in real time, resulting in delayed or interrupted dissemination of warning information.

Additional threats include political or legal restrictions affecting international information sharing, the increasing risk of collisions with space debris, and dependence on foreign launch providers, which may introduce schedule uncertainties and supply chain vulnerabilities.

The interaction analysis between weaknesses and threats highlights that operational limitations—including limited maintenance capabilities, satellite degradation, the complexity of multi-satellite data integration, and reliance on uninterrupted inter-satellite communication links—significantly increase the system's exposure to external risks. These factors may reduce mission resilience in the event of cyber incidents, hostile electronic interference, ISL disruption, or failures caused by the orbital environment. Consequently, mitigating these weaknesses through appropriate system design, communication redundancy, robust cybersecurity measures, reliable operational procedures, and careful mission planning should remain a priority

throughout the system development process.

9.5 Strategy

The overall strategic assessment clearly indicates that the project should adopt an **Attack (SO) Strategy**. This strategy achieved the highest overall score, with the greatest number of interactions and the highest weighted value among all strategic options. The results indicate that the project's strengths significantly outweigh its weaknesses and can be effectively leveraged to exploit favourable external opportunities.

The system's unique regional capability, combined with the growing demand for missile early-warning systems supporting civil protection and the availability of institutional funding programmes, creates favourable conditions for successful implementation, future expansion, and long-term operational deployment.

Although the **Retreat (WT) Strategy** achieved the second-highest score, reflecting the presence of significant technical and operational risks, these challenges do not outweigh the project's strategic advantages. Instead, they highlight the importance of proactive risk mitigation, particularly in the areas of cybersecurity, satellite reliability, launch independence, communication resilience, and long-term system maintainability.

9.6 Conclusions

The SWOT analysis demonstrates that the proposed hybrid GEO-LEO satellite constellation represents a promising and strategically justified investment in regional civil protection and missile early-warning capabilities. Its principal advantages-including early threat detection, high operational reliability, continuous availability, and the ability to archive and analyse historical event data-provide a strong foundation for addressing the growing resilience requirements of the Baltic region.

Nevertheless, the long-term success and operational effectiveness of the EWAS mission will depend on the effective management of technical, operational, and security-related risks. Particular attention should be devoted to strengthening cybersecurity, improving communication resilience, ensuring system redundancy, and maintaining the long-term sustainability of the satellite constellation. Addressing these challenges during the design, implementation, and operational phases will significantly increase the mission's reliability and overall strategic value.

10 Work Breakdown Structure

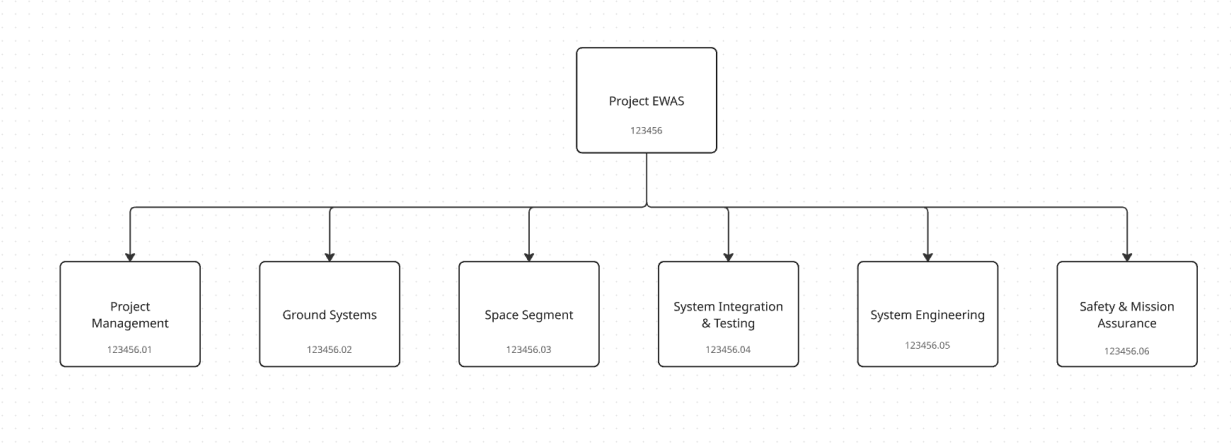


Figure 10.1: Work Breakdown Structure - Lvl 1 and Lvl 2

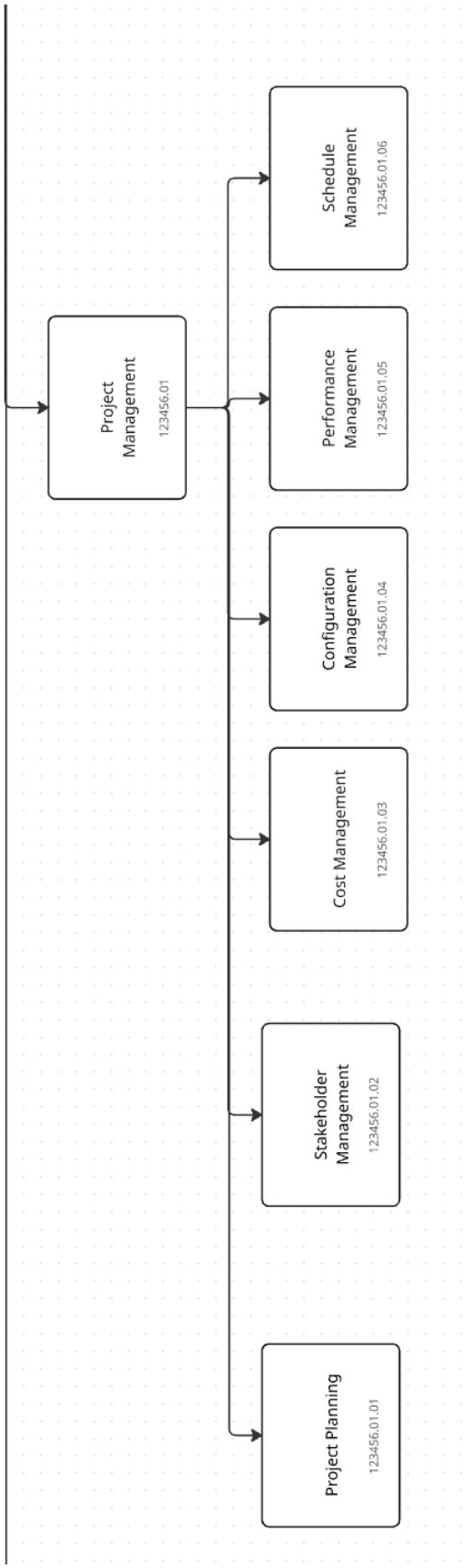


Figure 10.2: Project Management WBS extension

Element 123456.01 - Project Management: The management products required to plan, control, and monitor project execution. Includes project planning documents, schedules, cost baselines, performance measurement products, configuration management records, stakeholder management data, and project reporting products.

Element 123456.01.01 - Project Planning: Products defining the overall planning framework for project execution. Includes project management plans and work breakdown structure definitions used to organize and control project scope.
Deliverables:

- Project Management Plan
- Project Scope and Objectives Document
- Project Schedule and Milestone Plan
- Project Risk Management Plan
- Work Breakdown Structure Diagram
- WBS Dictionary
- WBS Coding Structure

Element 123456.01.02 - Stakeholder Management: Products supporting identification, communication, and monitoring of stakeholder interactions. Includes stakeholder data, communication plans, and status reporting.
Deliverables:

- Stakeholder Register
- Stakeholder Roles and Responsibilities Matrix
- Project Communication Plan
- Project Communication Matrix
- Stakeholder Communication Schedule
- Periodic Project Progress Reports
- Project Issues and Actions Log
- Milestone Review Reports

Element 123456.01.03 - Cost Management: Products used to estimate, define, and control project costs. Includes cost baselines and cost estimation data.
Deliverables:

- Project Cost Baseline
- Time-Phased Budget Allocation
- Preliminary Project Cost Estimate
- Cost Estimation Assumptions and Basis
- Cost Breakdown by Major Project Elements

Element 123456.01.04 - Configuration Management: Products used to control and track changes to project deliverables and documentation. Includes change records, configuration baselines, and control authority decisions.
Deliverables:

- Change Request Log
- Change Impact Assessment Records
- Change Approval and Rejection Records
- Configuration Control Board Meeting Minutes
- Configuration Change Decision Log
- Approved Document Baseline
- System Configuration Baseline
- Configuration Item Register

Element 123456.01.05 - Performance Management: Products used to measure and monitor project performance against cost, schedule, and technical baselines. Includes performance baselines and earned value data products.
Deliverables:

- Performance Measurement Baseline
- Project Performance Indicators and Metrics
- Earned Value Management Reports

- Cost and Schedule Variance Reports
- Project Performance Trend Analysis

Element 123456.01.06 - Schedule Management: Products used to plan and control project timelines. Includes schedules and approved schedule baselines for tracking progress. *Deliverables:*

- Approved Project Schedule Baseline
- Project Milestone Schedule
- Integrated Master Schedule (IMS)
- Project Activity Dependency Network
- Critical Path Analysis Report

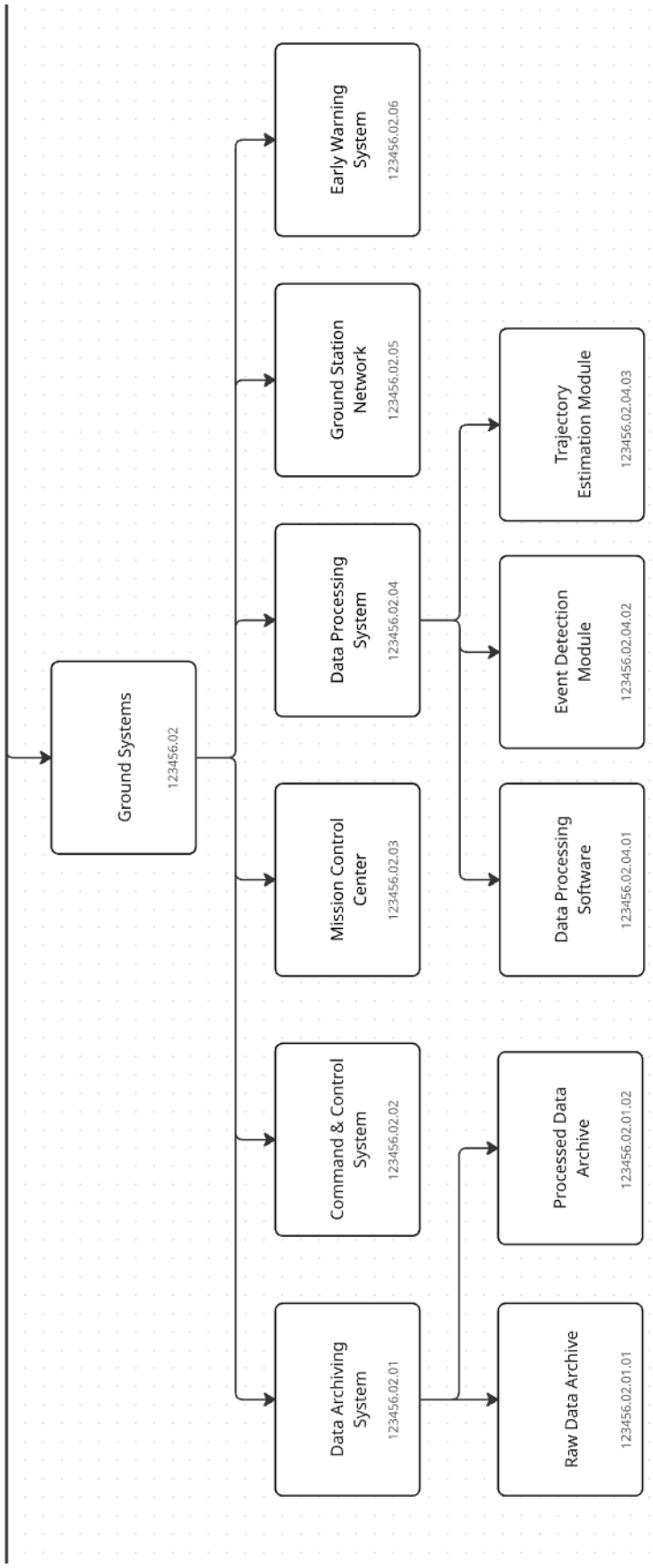


Figure 10.3: Ground Systems extension

Element 123456.02 - Ground Systems: The systems, facilities, software, and networks required to operate the space segment and support mission execution. Includes command and control, data processing, data archiving, ground station infrastructure, and mission-specific operational systems such as early warning capabilities.
Element 123456.02.01 - Data Archiving System: System responsible for storage, management, and retrieval of mission data. Includes infrastructure for handling raw and processed data products generated during mission operations.
Element 123456.02.01.01 - Raw Data Archive: Repository for unprocessed mission data received from spacecraft systems and ground stations. Includes storage of telemetry, sensor outputs, and initial acquisition data.
Element 123456.02.01.02 - Processed Data Archive: Repository for processed and analyzed data products derived from raw mission data. Includes refined datasets used for mission analysis and operational decision-making.
Element 123456.02.02 - Command & Control System: System responsible for generation, transmission, and management of commands sent to the spacecraft, as well as monitoring and control of spacecraft operations. Includes command interfaces and control software.
Element 123456.02.03 - Mission Control Center: Facility and associated systems used for monitoring spacecraft status, executing mission operations, and coordinating ground system activities. Includes operator interfaces, monitoring systems, and operational control tools.
Element 123456.02.04 - Data Processing System: System responsible for processing, transforming, and analyzing mission data received from the spacecraft. Includes software and algorithms required to extract meaningful information for mission operations.
Element 123456.02.04.01 - Data Processing Software: Software responsible for ingestion, conditioning, and transformation of raw data into formats suitable for analysis. Includes preprocessing, filtering, and data preparation functions.
Element 123456.02.04.02 - Event Detection Module: Software module responsible for identifying significant events within processed data, such as launch detection or anomaly identification.
Element 123456.02.04.03 - Trajectory Estimation Module: Software module responsible for estimating object trajectories based on processed data inputs. Includes algorithms for calculating position, velocity, and predicted paths.
Element 123456.02.05 - Ground Station Network: Network of ground-based stations used to communicate with the spacecraft. Includes antennas, communication links, and supporting infrastructure for telemetry reception and command transmission.
Element 123456.02.06 - Early Warning System: System responsible for generating and disseminating alerts based on analyzed mission data. Includes components for event classification, alert generation, and communication of warning information to external users or systems.

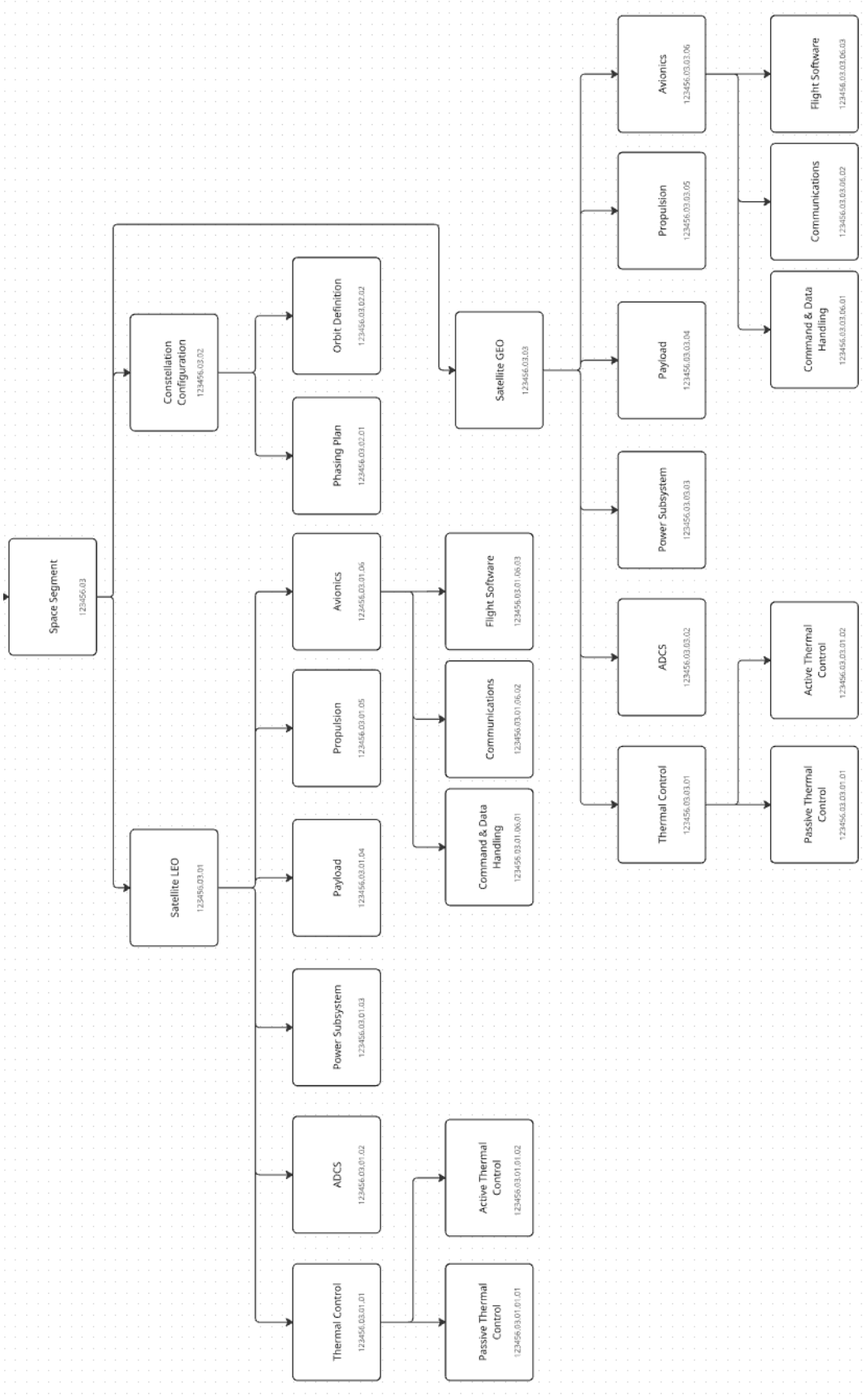


Figure 10.4: Space Segment extension

Element 123456.03 - Space Segment: The hardware and software elements required to perform the space-based mission functions. This includes the satellite system and constellation configuration defining orbital parameters and satellite distribution.
Element 123456.03.01 - Satellite: The flight system consisting of all onboard subsystems required to support mission operations in orbit. Includes payload, spacecraft bus subsystems, avionics, propulsion, and flight software.
Element 123456.03.01.01 - Thermal Control: Subsystem responsible for maintaining spacecraft temperature within operational limits. Includes passive and active thermal control components and associated hardware.
Element 123456.03.01.01.01 - Passive Thermal Control: Subsystem components that maintain spacecraft temperature within acceptable limits without active power consumption. Includes thermal coatings, insulation (MLI), radiators, and structural thermal design features.
Element 123456.03.01.01.02 - Active Thermal Control: Subsystem components that regulate spacecraft temperature using powered devices. Includes heaters, thermostats, fluid loops, heat pipes, and associated control electronics.
Element 123456.03.01.02 - Attitude Determination and Control System (ADCS): Subsystem providing spacecraft orientation determination and control. Includes sensors, actuators, and control algorithms required for pointing and stabilization.
Element 123456.03.01.03 - Power Subsystem: Subsystem responsible for generation, storage, conditioning, and distribution of electrical power, including solar arrays, batteries, and power management electronics.
Element 123456.03.01.04 - Payload: Mission-specific instruments and sensors responsible for detection and data acquisition required to fulfill mission objectives.
Element 123456.03.01.05 - Propulsion Subsystem: Subsystem providing thrust for orbit insertion, maintenance, and maneuvering, including thrusters, propellant storage, and associated control components.
Element 123456.03.01.06 - Avionics: Onboard electronic systems responsible for command, data processing, communication, and software execution.
Element 123456.03.01.06.01 - Command & Data Handling (C&DH): Subsystem responsible for onboard data processing, command execution, and data management. Includes onboard computers, data buses, memory units, and associated control software required for spacecraft operations.
Element 123456.03.01.06.02 - Communications: Subsystem responsible for transmission and reception of telemetry, tracking, and command (TT&C) data between the spacecraft and ground systems. Includes antennas, transceivers, RF components, and communication protocols.
Element 123456.03.01.06.03 - Flight Software: Software executed onboard the spacecraft to control system functions, manage subsystem operations, process data, and execute mission tasks. Includes control algorithms, fault management, and operational logic.
Element 123456.03.02 - Constellation Configuration: Definition of the orbital architecture and spatial distribution of satellites within the constellation, including orbital parameters and phasing.
Element 123456.03.02.01 - Constellation Phasing Plan: The definition of satellite distribution and relative positioning within orbital planes to ensure required coverage and

mission performance.

Element 123456.03.02.02 - Orbit Definition: The specification of orbital parameters including altitude, inclination, eccentricity, and orbital plane configuration.

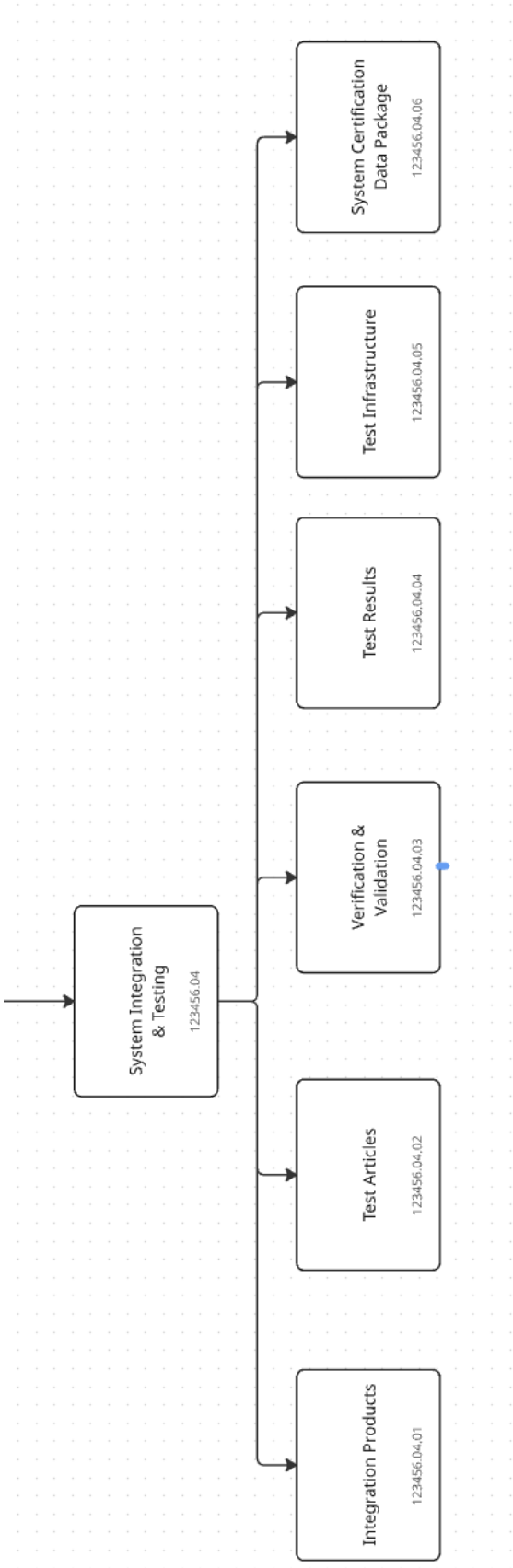


Figure 10.5: System Integration & Testing extension

Element 123456.04 - Systems Integration and Testing: This element includes the products, data, and supporting infrastructure required to integrate system components and verify system performance against requirements.
Element 123456.04.01 - Integration Products: Integrated hardware and software system configurations resulting from assembly and integration activities. Includes system-level configurations combining space, ground, and interface elements required for testing and verification.
Element 123456.04.02 - Test Articles: Physical system models and configurations used for verification and testing. Includes engineering, qualification, and flight models representing different stages of system maturity.
Element 123456.04.02.01 - Qualification Model: A representative model used to demonstrate compliance with environmental and performance requirements under qualification test conditions.
Element 123456.04.02.02 - Engineering Model: A non-flight configuration used for design verification, integration testing, and functional validation during development.
Element 123456.04.02.03 - Flight Model: The flight-ready system configuration intended for operational deployment and mission execution.
Element 123456.04.03 - Verification & Validation Products: Documentation and data products demonstrating compliance with system requirements and validation of mission objectives. Includes traceability matrices and validation reports.
Element 123456.04.03.01 - Verification Matrix: A structured data product mapping system requirements to verification methods and test results, providing evidence of requirement compliance.
Element 123456.04.03.02 - Validation Reports: Documentation confirming that the system meets intended operational goals and mission objectives under realistic conditions.
Element 123456.04.04 - Test Results: Documented outcomes of system-level and subsystem-level testing activities. Includes environmental, functional, and performance test results demonstrating system behavior.
Element 123456.04.05 - Test Infrastructure: Facilities, equipment, and tools required to support system integration and testing. Includes ground support equipment, test benches, and simulation environments.
Element 123456.04.05.01 - Test Benches: Dedicated test setups configured to evaluate specific subsystems or functions. Includes hardware and software environments supporting functional and integration testing.
Element 123456.04.05.02 - Ground Support Equipment: Equipment used to support testing, integration, and operation of the system from the ground. Includes electrical and mechanical interfaces, control systems, and support hardware.
Element 123456.04.05.03 - Simulators: Software or hardware systems used to emulate spacecraft, environment, or mission conditions for testing and validation purposes.
Element 123456.04.06 - System Certification Data Package: A compiled set of verification, validation, and test data used to demonstrate system compliance with requirements and readiness for operational use. Includes supporting evidence from test results, analysis, and verification activities.

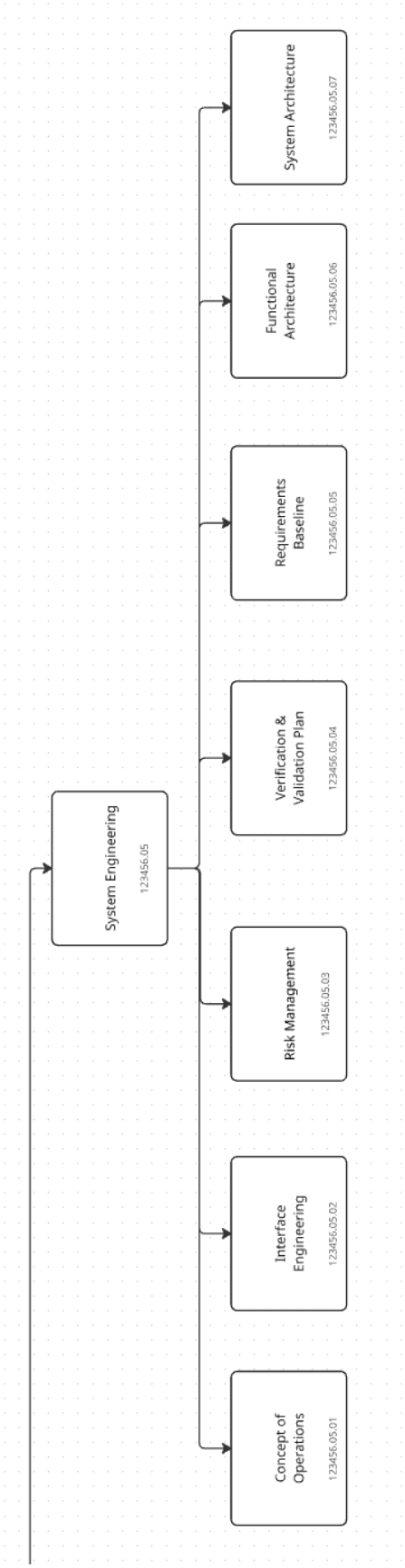


Figure 10.6: System Engineering extension

Element 123456.05 - Systems Engineering: The products required to define, integrate, and verify the technical baseline of the system. Includes requirements documentation, system architecture descriptions, interface control documents, verification and validation plans, risk management products, and concept of operations documentation.
Element 123456.05.01 - Concept of Operations Document: A document describing how the system is intended to operate in its mission environment. Includes operational scenarios, system interactions, and end-to-end mission workflows.
Element 123456.05.02 - Interface Control Documents: Documentation defining interfaces between system elements. Includes interface specifications for data exchange, external systems, and segment-level interactions.
Element 123456.05.02.01 - Data Exchange Interface Control Document: Documentation defining data formats, protocols, and interfaces for information exchange between system components.
Element 123456.05.02.02 - External Systems Interface Control Document: Documentation defining interfaces between the system and external entities or partner systems.
Element 123456.05.02.03 - Ground Segment Interface Control Document: Documentation defining interfaces within the ground system and between ground and other system elements.
Element 123456.05.02.04 - Space Segment Interface Control Document: Documentation defining interfaces within the space segment and between spacecraft subsystems.
Element 123456.05.03 - Risk Management: Products used to identify, analyze, track, and mitigate project risks. Includes risk register, risk assessment results, mitigation plans, and risk reporting products.
Element 123456.05.03.03 - Risk Register: A structured repository of identified risks, including descriptions, probabilities, impacts, and status.
Element 123456.05.03.04 - Risk Assessment Results: Documented evaluation of risks, including probability and impact analysis results.
Element 123456.05.03.04.01 - Probability Scale Definition: Definition of probability levels used to assess risks consistently across the project.
Element 123456.05.03.04.02 - Impact Assessment Results: Evaluation of potential consequences associated with identified risks.
Element 123456.05.03.02 - Risk Mitigation Plans: Documentation defining strategies and actions to reduce risk probability and impact, including contingency approaches.
Element 123456.05.03.01 - Risk Status Reports: Reports providing updates on risk status, mitigation progress, and risk trends.
Element 123456.05.03.01.01 - Risk Tracking Reports: Data products monitoring changes in risk status over time.
Element 123456.05.03.01.02 - Risk Review Reports: Documentation of periodic risk reviews and associated decisions.
Element 123456.05.04 - Verification & Validation Plan: Documentation defining the methods, processes, and criteria used to verify and validate system performance against requirements.
Element 123456.05.05 - Requirements Baseline: The complete set of approved system and mission requirements, including traceability and supporting documentation.
Element 123456.05.05.01 - Requirements Traceability Matrix: A structured map-

ping between requirements and their verification methods and implementation elements.
Element 123456.05.05.02 - System Requirements Document: Documentation defining technical requirements for system design and implementation.
Element 123456.05.05.03 - Mission Requirements Document: Documentation defining high-level mission objectives and constraints.
Element 123456.05.06 - Functional Architecture Model: Representation of system functions and their relationships, describing how system processes are organized and interact.
Element 123456.05.07 - System Architecture Description: Documentation describing the physical and logical structure of the system, including components, interfaces, and interactions.

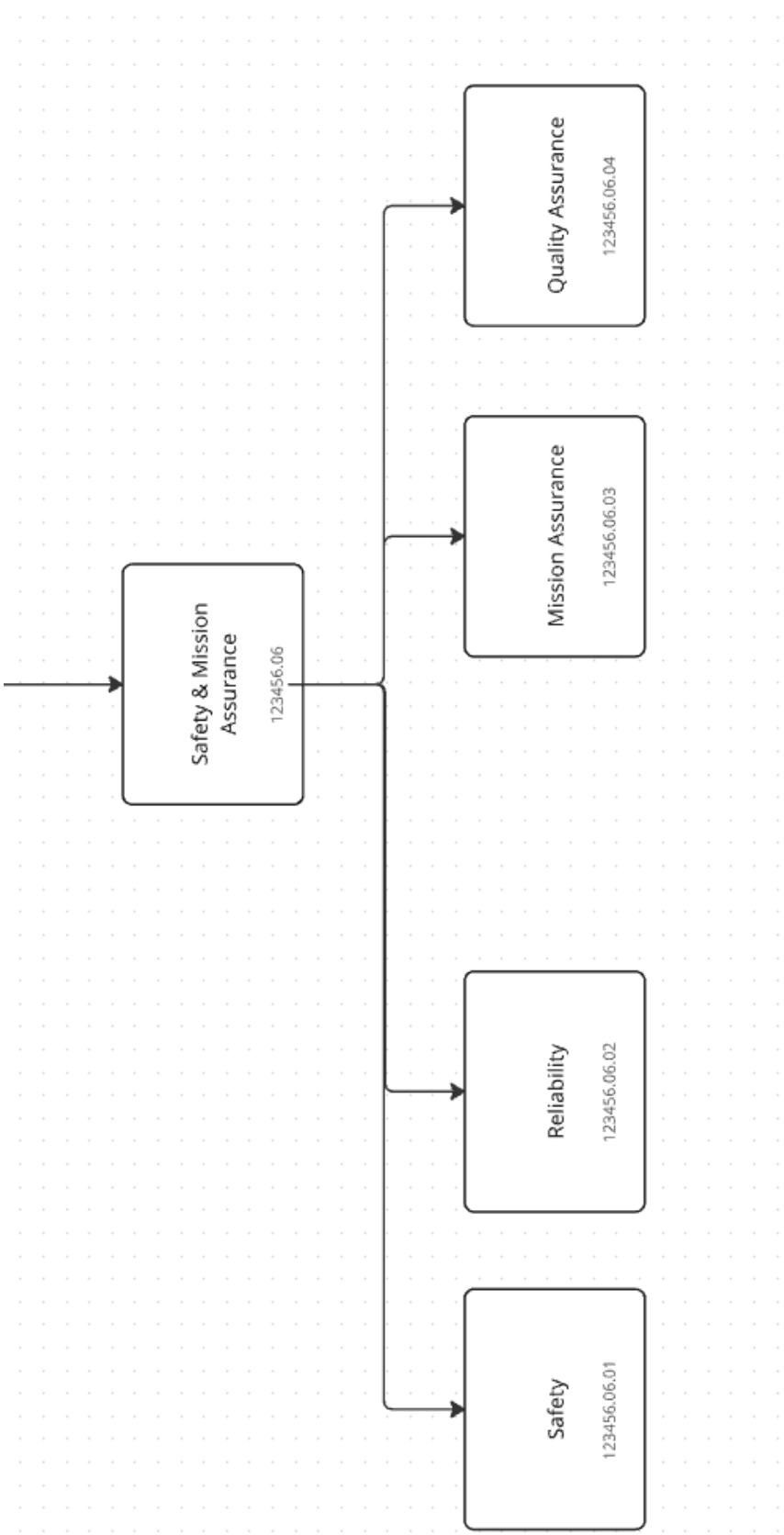


Figure 10.7: Safety & Mission Assurance extension

Element 123456.06 - Safety & Mission Assurance: The products required to ensure system safety, reliability, quality, and mission success. Includes safety data products, reliability analysis results, quality assurance records, and certification documentation required to demonstrate system compliance and readiness.
Element 123456.06.01 - Safety: Products related to identification, evaluation, and control of hazards to ensure system safety. Includes hazard analysis data, safety tracking systems, and safety review documentation.
Element 123456.06.01.01 - Hazard Analysis Report: Documentation identifying potential hazards, their causes, and associated mitigation measures to ensure safe system operation.
Element 123456.06.01.02 - Safety Review Data Package: A collection of safety- related documentation supporting safety reviews, including hazard analysis and compliance evidence.
Element 123456.06.01.03 - Hazard Tracking System: A system used to monitor and manage identified hazards, including status, mitigation actions, and closure records.
Element 123456.06.02 - Reliability: Products related to assessment and prediction of system reliability. Includes reliability analysis results, failure data, and modeling outputs.
Element 123456.06.02.01 - Reliability Prediction: Analytical results estimating system reliability metrics such as failure rates and mean time between failures.
Element 123456.06.02.02 - FMEA and Fault Tree Analysis Results: Analysis results identifying failure modes, their effects, and system-level fault propagation.
Element 123456.06.02.03 - Failure Reports: Documentation describing system failures, anomalies, and associated root cause analysis.
Element 123456.06.03 - Mission Assurance: Products ensuring that the system meets mission success criteria and operational requirements. Includes certification documentation, compliance evidence, and readiness support materials.
Element 123456.06.03.01 - Flight Readiness Review Data Package: Documentation supporting the evaluation of system readiness for operational deployment.
Element 123456.06.03.02 - Compliance Verification Evidence: Documented evidence demonstrating that system requirements and standards have been met.
Element 123456.06.03.03 - Final Certification Package: The approved documentation set confirming system compliance and readiness for operational use.
Element 123456.06.04 - Quality Assurance: Products used to ensure that system development and manufacturing comply with defined standards. Includes inspection records, audit reports, and nonconformance documentation.
Element 123456.06.04.01 - Inspection Record: Documentation of inspections performed on hardware, software, or processes to verify compliance with requirements.
Element 123456.06.04.02 - Quality Assurance Audit Reports: Reports documenting results of audits conducted to verify compliance with quality standards and procedures.
Element 123456.06.04.03 - Nonconformance Reports: Documentation of deviations from requirements, including descriptions, impacts, and corrective actions.

11 Overall Schedule

This chapter presents the planned sequence of activities for preparing the EWAS final report. The work is divided into three phases so that the mission concept is established before the technical architecture is defined and the complete document is integrated.

11.1 Documentation Schedule

Table 11.1 summarizes the sequence of documentation activities, identifies the team member responsible for each deliverable, and specifies the corresponding completion deadline. The phased structure reflects the dependencies between mission definition, technical development, and final project integration.

Table 11.1: Documentation schedule and task ownership

Deliverable	Owner	Deadline
Phase 1 — Mission and Concept (April)		
Mission definition: need, purpose, objectives, scope, and success criteria	Grzegorz	15 April
Use scenarios and CONOPS	Michał	15 April
Proposed solution concept	Krzysztof	30 April
Phase 2 — System Architecture and Requirements (May)		
Overall system requirements: draft completion	Ludwik	15 May
Functional analysis: draft completion	Sergiusz	15 May
Subsystem design	Jan	31 May
Phase 3 — Project Planning and Finalization (June)		
Stakeholder analysis and risk/SWOT analysis	Julia and Ola	15 June
WBS	Maciej	15 June

11.2 Key Milestones

The key milestones in Table 11.2 define the principal review and delivery points of the documentation process. Each milestone includes a completion criterion that can be used to assess whether the project is ready to proceed to the next stage.

Date	Milestone	Completion Criterion
15 April	Mission baseline review	The mission definition, use scenarios, and CONOPS are sufficiently mature to support technical requirements development.
31 May	Technical baseline review	The proposed solution, system requirements, functional analysis, and subsystem design are complete and mutually consistent.
30 June	Final report delivery	All technical and project-planning sections are integrated, reviewed, and prepared for submission.

Table 11.2: Key documentation milestones